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PANEL-B (B2)
ROLL No. → 57
MPJ

LAB ASSIGNMENT-01

- **Aim:** Write a Java Program to showcase classes, object creation, methods, constructors using student information.
- **Objectives:**
 - i) To study classes, objects & creation of objects of a class.
 - ii) To study methods, return a value from the method, & pass parameters to the methods.
 - iii) To study constructors & use of constructors.
 - iv) To study different access specifiers in Java & their use for encapsulation.

- **Theory:**

- i) Define what is class & object with an example.

→ **Class:** A class is a blueprint / template to create objects.

It defines variables & methods of an object. It is a user-defined data type that groups data members & member functions into a single unit.

Eg:

```
class Student {  
    int id;  
    String name;  
}
```

Object: It is an instance of a class that represents a real-world entity & occupies

memory. It is a runtime entity created from a class.

Eg-

```
public class Main {  
    public static void main( String[] args ) {  
        Student s1 = new Student();  
        s1.id = 10;  
        s1.name = "Arushi";  
        s1.display();  
    }  
}
```

ii) Define the syntax to define methods in a class & give examples for parameter passing & returning a value from methods.

→ Basic syntax:

```
return-type method-name( parameter-list ) {  
    // method body  
    return value; }  
}
```

Eg 1: Method with parameters & no return value :

```
class Calculator {  
    void add( int a, int b ) {  
        System.out.println( a+b );  
    }  
}
```

Method call :

```
Calculator c = new Calculator();  
c.add( 10, 20 );
```

with

Eg 2: Method parameters & return value :

```
class Calculator {  
    int multiply( int a, int b ) {  
        return a * b; }  
}
```

Method Call :

```
Calculator c = new Calculator();  
int result = c.multiply();  
System.out.println( result );
```

iii) Write about the use of constructor with example.

→ A constructor is a special method used to initialize objects. It helps assigning default values, has the same name as class & no return type.

eg → `Student (int i, String n) { // given class Student.`

`id = i;`

`name = n; }`

• FAQs :-

Q.1 What is a class & object?

Ans Class : A class is a blueprint or template used to create objects. It defines the properties and methods of an object.

Object : An object is an instance of a class. It represents a real-world entity and occupies memory at runtime.

Q.2 What is a constructor in Java?

A constructor is a special method used to initialize objects.

→ It has the same name as the class & is automatically invoked when an object is created.

Key points :-

i) It has no return type.

ii) It is called automatically.

iii) It is used to initialize data members.

Q.3 What is the default access specifier & what is its scope?

Ans: In Java, when no access modifier is specified for a class member assigned to the default access specifier automatically.

Scope :-

- i) Members are accessible within the same package.
- ii) Members aren't accessible outside the package.
- iii) It helps in package-level data hiding.
- iv) It's also known as package private access.

Q.4 What are the different access specifiers?

Ans Java has 4 types of access specifiers :-

- i) Private: Accessible only within the same class, used for encapsulation.
- ii) Default: Accessible within the same package, applied when no access specifier is specified.
- iii) Protected: Accessible within the same package, also accessible to outside the package.
- iv) Public: Accessible from ~~any~~ anywhere, used for methods or classes meant for global access.